ECOR2050: Design and Analysis of Engineering Experiments

Assignment 1: Due September 30, 2026

Carleton UNIVERSITY
Canada's Capital University

Instructions:

1. Solutions should be compiled in a single file and submitted to Brightspace.
2. Answer all parts of each question.
3. Show all manual equations, calculations, and units.
4. Plots must be labelled, clear, and legible.
5. Plots must be completed in software.
6. Every solution must include a concluding statement that interprets the numeric result. Correct numeric results without a concluding statement will lose marks.

Q1. The following table contains price data in dollars for textbooks in a major university's bookstore.

29.70	19.95	29.95	11.25	27.67
24.70	19.50	124.80	89.73	50.95
29.65	11.95	152.48	183.75	18.75

(a) Calculate the mean and standard deviation of the data. **[3 marks]**

(b) Calculate the Pearson coefficient of skewness of the data. Which type of Pearson coefficient did you use and why? **[3 marks]**

(c) In your opinion, which metric of central location is best at describing the "middle" of this dataset? Justify your opinion. **[2 marks]**

(d) Calculate the 65th percentile of the sample. What is the interpretation of this number? **[3 marks]**

(e) Identify which numbers in the data (if any) are outliers. Identify which numbers (if any) are extreme outliers. Show all calculations to support your answer. **[7 marks]**

(f) Create a cumulative relative frequency plot of the data. Use it to estimate the 65th percentile of the sample. Compare your result to part (d). **[6 marks]**

(g) Create a box-and-whisker plot of the data. Label all key values on the plot. Does the plot agree with your answer to part (e)? **[6 marks]**